

OHBM 2026 Hackathon project: BIDS-layerfMRI

Description

High-resolution layer-fMRI is a rapidly growing field, with more than 350 studies published to date. As the community expands, there is an increasing need for standardized approaches to organize, share, and reuse layer-fMRI datasets. The Brain Imaging Data Structure (BIDS) provides a widely adopted framework for standardizing file naming conventions and dataset organization, facilitating data sharing and reproducibility. However, clear guidelines for organizing and sharing both source and processed layer-fMRI data within the BIDS framework are still lacking.

Building on previous efforts initiated by Renzo Huber during a hackathon in 2022, this project aims to help address this gap by:

- Gathering feedback from layer-fMRI experts through interviews and community surveys: <https://forms.gle/z4N8eTTMXAA9mbST7>
- Providing an example layer-fMRI dataset that can serve as a practical reference for data organization and sharing.
- Submitting a pull request to the BIDS community to incorporate these recommendations into future standards.

Based on discussions with the community during the OHBM BrainHack (2026), we propose the following structure as “initial prototype” with the minimum requirement for organizing layer-fMRI datasets.

Dataset_layer-fMRI_example

Dataset_description.json

```
├─ sub-02
|   └─ sess-02
|       └─ func
|           └─ sub-02_sess-04_task-movie_run-01_bold.nii.gz
|           └─ anat
|           └─ sub-02_sess-01_run-01_T1w.nii.gz
├─ derivative
|   └─ anat
└─ sub-02_desc-preproc_T1w.nii.gz
```

```

|   └─ sub-02_seg-braintissues_dseg.nii.gz
|   └─ braintissues_dseg.tsv
|   └─ sub-02_space-NATIVE_atlas-Glasser.nii.gz
|   └─ atlas-Glasser_description.json
|   └─ sub-02_seg-depth_method-equivol.nii.gz
|   └─ depth_description.json
|   └─ sub-02_seg-layer_method-equivol.nii.gz
|   └─ layer_dseg.tsv
|   └─ sub-02_roi-V1_dseg.nii.gz
|   └─ roi_dseg.tsv
|   └─ func
|   └─
sub-02_sess-04_task-movie_run-01_space-NATIVE_desc-prepro_bold.nii.gz
|   └─
sub-02_sess-04_task-movie_run-01_space-NATIVE_stat-mean_bold.nii.gz
|   └─
sub-02_sess-04_task-movie_run-01_space-NATIVE_stat-tsnr_bold.nii.gz

```

The source data organization follows the current BIDS specification for MRI and fMRI datasets. However, for the derivative data organization some of the file names and conventions included in this proposal are not currently part of the official BIDS standard. Rather, they represent a first community-driven attempt intended to facilitate the sharing and reproducibility of layer-fMRI data while broader standards are being developed.

For describing geometrical cortical layers, we propose the following file naming convention:

```

...

sub-02_seg-depth_method-equivol.nii.gz

sub-02_seg-layer_method-equivol.nii.gz

...

```

In both cases, it is important to specify the method used to compute cortical depth, such as **equidistant** or **equivolume** layering. The ``depth`` file contains a continuous coordinate representation of cortical depth, typically ranging from 0 to 1 across the cortical ribbon. This file serves as the basis from which discrete cortical layers are derived.

In the example above, both outputs were generated using LayNii (v2.10.0).

The accompanying `depth_description.json` file specifies how the depth coordinate is defined, including its orientation with respect to the cerebrospinal fluid (CSF) and white matter boundaries.

0 for boundary between WM and GM, 1 for boundary between GM and SCF.

Similarly, the `layer_dseg.tsv` file describes the discrete geometrical layers and their corresponding positions within the cortical ribbon.

`layer_dseg`

index	name
1	layer 1 (deep)
2	layer 2
3	layer 3
4	layer 4
5	layer 5
6	layer 6
7	layer 7
8	layer 8
9	layer 9
10	layer 10
11	layer 11 (superficial)

In some cases, layer-fMRI researchers may wish to restrict analyses to a specific brain region or a subject-specific subregion thereof. For this purpose, we propose the use of ``roi`` files. While these files are conceptually similar to atlases, the term **region of interest (ROI)** may be more appropriate and intuitive for layer-fMRI analyses, where masks are often generated for a particular cortical area or a subset of that area rather than representing a whole-brain parcellation.

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Bollmann, Marco Barilari, Remi Gau, Avinash Kalyani, and Omer Faruk Gulban for their discussions during the OHBM Brainhack 2026. This example dataset is a part of Kenshu dataset: <https://openneuro.org/datasets/ds003216>

We welcome feedback from the community. If you have comments, suggestions, or concerns, please complete the survey: <https://forms.gle/z4N8eTTMXAA9mbST7>